

Functions

1. Names of parameters in a function prototype have to match the names given in the function definition. TRUE/FALSE
2. Write a function named `int zeroCheck(int a, int b, int c)`; that is given three integers, and returns 1 if any of the integers is 0, otherwise it returns 0.

```
int zeroCheck(int a, int b, int c)
{
    if(a==0 || b==0 || c==0)
        return 1;
    else
        return 0;
}
```

3. What is the output for the following code:

```
#include <stdio.h>
int what(int a, int n)
{
    if(n == 0)
        return 1;
    else if(n % 2)
        return a * what(a * a, n / 2);
    else
        return what(a * a, n / 2);
}
int main()
{
    int a = 3, b = 5;
    printf("%d\n", what(a, b));
}
```

Output:

243

4. Which of the following would be valid prototypes for a function that returns nothing and has one double parameter?

- a. `void f(double x);`
- b. `void f(double);`
- c. `void f(x);`
- d. `f(double x);`

5. Write a program to find the number of digits in an integer using recursion.

`int length(int n);`

`length(892)` will return 3

`length(3452)` will return 4.

```
int len(int n)
{
    if(n==0)
        return 0;
    else
        return 1+len(n/10);
}
```